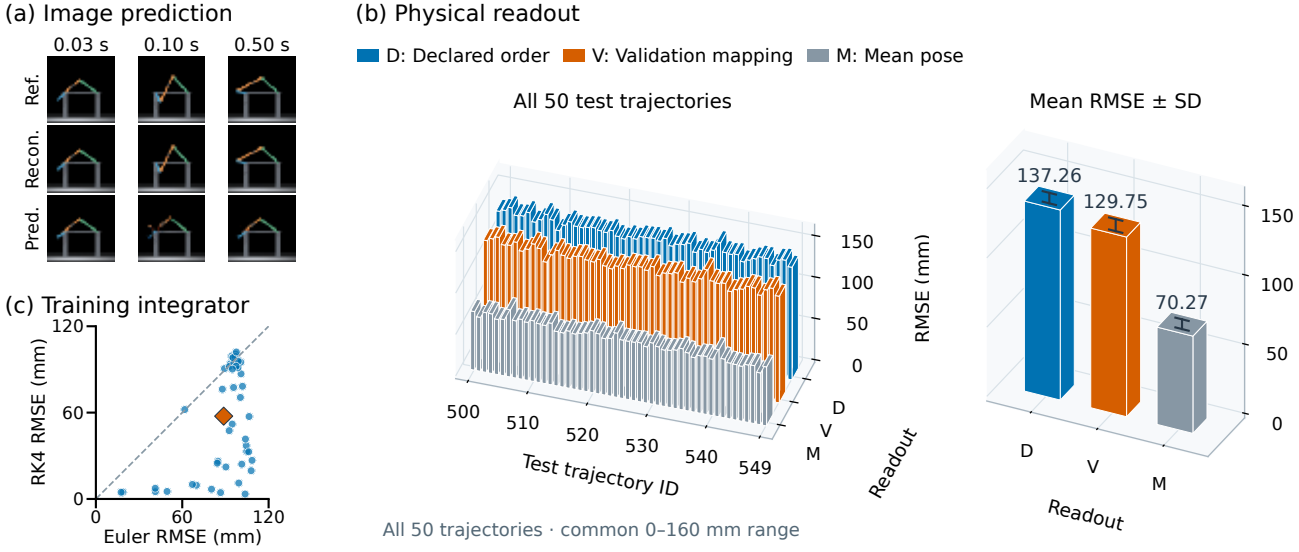
(a) Image prediction
0.03 s
0.10 s
0.50 s
Ref.
Recon.
Pred.
(b) Physical readout
D: Declared order
V: Validation mapping
M: Mean pose
All 50 test trajectories
Test trajectory ID
500
510
520
530
540
549
RMSE (mm)
0
50
100
150
D
V
M
Readout
Mean RMSE ± SD
137.26
129.75
70.27
(c) Training integrator
RK4 RMSE (mm)
Euler RMSE (mm)
0
60
120
All 50 trajectories · common 0–160 mm range